

### **\*OSI Layers\***

OSI stands for open systems interconnection . OSI model contains 7 layers

Application layer - application layer is user interface layer . it works on some protocols like HTTP, HTTPS, SMTP

Presentation layer - presentation layer handles compression, encryption and decryption of data.

session layer - session layer coordinates and terminates conversation between the applications. it maintain sessions between the client and server.

transport layer - It is known as heart of OSI model. it is responsible for transferring the data across the networks . it uses protocols like TCP and UDP. here the tcp header is added and data transfers into segments.

Network layer - the primary function of network layer is to move data into another layer. this is where ip address are applied for routing purposes. here segments are transfers into packets by adding ip headers.

Datalink layer - this layer handles moving data into "IN AND OUT" of physical link in a network . it assures that error free data is been transferred into frames by adding ethernet header with mac address.

physical layer - It handles raw data and responsible for sending computer bits from one device to another device across the network.

### **\*Difference with TCP/IP Model\***

OSI model is a theoretical model and TCP/Ip model is a real time implementation model. TCP/Ip have four layers and OSI have seven layers. TCP/Ip is protocol dependent model and OSI is protocol independent model.

### **\*Network Topology When Typing google.com\***

when we browse for www.google.com in web browser . it generates a DNS query. it checks in browsing cache . if domain is not in cache the query is forwarded to recursive DNS server. once the DNS response was received then TCP connection is established by TCP-three way handshake. if connection is via port 80 then HTTP comes into picture. if connection is port 443 then SSL handshake happens to establish a secure connection between client and server . once the secure connection is established then web page will be loaded.

### **\*DNS Records\***

there are many DNS records for example

-A record = which maintain IPv4 address records

-AAAA records = which maintain IPv6 address records.

-TX records = which maintain text records.

-MX records = which maintain mail server details.

-PTR(pointer records) = Used for mapping IP address to domain names.

-SOA records = (Start of authority ) it maintains authoritative server details. it determines the TTL value.

### **\*DHCP DORA Process\***

DHCP stands for dynamic host configuration protocol. it is used for assigning IP address dynamically for client or for assigning ip address automatically.

it follows a process called DORA

DORA process is

Discover - In discover message client broadcast a message with source Ip 0:0:0:0 , destination Ip 255:255:255:255 into the network source mac as client mac and destination mac as FFFFFFFF.

Offer -DHCP server provide an offer with destination mac as client mac. source mac as server mac address . source Ip address as server ip address .destination Ip address as 255:255:255:255

Request - when client receives the offer message from the DHCP server .it broadcast a gracious ARP to check in order to any host in the network with the same Ip address . if there is no response from ARP then request message is broadcasted to the server showing the acceptance of Ip address offered .

Acknowledgement - The DHCP server acknowledges the request made by the client by providing Ip address requested along with gateway Ip, subnet mask, DNS server Ip, lease times and few other attributes.

### **\*TCP Three-Way Handshake\***

In TCP three-way handshake client initiates the connection by sending a SYN packet to the server with following parameters-sequence number, window size, length, MSS, window scaling factor and sack premiant, then server sends a SYN-ACK message along with same parameters by adding acknowledged number. once client receives SYN-ACK message from the server then client acknowledges the SYN send by the server.

In TCP there are six well known flags

- synchronize
- Acknowledge
- push
- Urgent
- Reset
- Finish

### **\*HTTP Request Methods & Response Codes\***

Common HTTP methods include GET, POST, PUT, DELETE. Response codes are 1XX for information, 2XX for success, 3XX for redirection, 4XX for client error, 5XX for server error.

### **\*SSL Handshake\***

Client initiates SSL handshake by sending a "CLIENTHELLO" message to the server. This message includes TLS version, client supported cipher suits and random number along with SNI (server name index). In reply server sends "serverhello" message to client containing server SSL certificate, chosen cipher suits and random number. Client verifies the server SSL certificates with the certificate authority. Once certificate validation is done, client shares a premaster secret which is encrypted by public key and it can only be decrypted by private key of the server. Then both client and server generate session keys. After that client sends a finished message that is encrypted by session keys. A secure symmetric encryption is achieved. Then communication happens in a secured channel.

### **\*SAML Flows: In \*\*IDP-initiated flow and In \*\*SP-initiated flow\***

SAML stands for security assertion markup language. It is a protocol that allows users to access multiple applications with a single set of credentials.

In IDP-initiated flow

The user logs in to the identity provider. The identity provider initiates login by sending a SAML response to the service provider. The SAML response contains a SAML assertion that tells the service provider who the user is. The service provider validates the signature in the SAML response and identifies the user. The user is now logged in to the service provider.

In SP-initiated flow

The user tries to access the Service Provider. The service provider initiates login by sending a SAML request to the identity provider, asking it to authenticate the user. The identity provider sends the user to a login page. The user enters their identity provider login credentials and the identity provider authenticates the user. The identity provider now knows who the user is, so it sends a cryptographically signed SAML response to the service provider. The SAML response contains a SAML assertion that tells the service provider who the user is. The service provider validates the signature in the SAML response and identifies the user. The user is now logged in to the service provider and can access the protected resource.

### **\*IPSEC Versions & MTU/MTR/MSS\***

IPSEC has two modes: Transport mode (encrypts payload) and Tunnel mode (encrypts entire packet).

Tunnel mode - Protects data in network-to-network or site-to-site scenarios. It encapsulates and protects the entire IP packet-the payload including the original IP header and a new IP header (protects the entire IP payload including user data).

Transport mode - Protects data in host-to-host or end-to-end scenarios. In transport mode, IPsec protects the payload of the original IP datagram by excluding the IP header (only protects the upper-layer protocols of IP payload (user data)).

IPsec protocols AH and ESP can operate in either transport mode or tunnel mode.

In IPSEC configuration we have two phases. phases 1 and phase 2 .phase 1 is used for connection establishment. phase 2 is used for data transferring . in phase 1 we configure in two modes , one is main mode and another one is aggressive mode. phase 2 is for quick mode . In main mode we have six messages . in aggressive mode we have three messages . main mode is more secure then aggressive mode .

IPsec use port 500 .

MTU - The maximum transmission unit (MTU) of an interface tells Cisco IOS the largest IP packet that can be forwarded out on that interface.

MTR -(my trace route ) it is a tool that combines trace route and ping.

MSS -(maximum segment size) defines largest segment tcp can send.

#### **\*GRE/IPSEC Differences\***

GRE (Generic Routing Encapsulation) is a tunneling protocol that encapsulates packets, while IPSEC provides encryption and authentication for securing IP packets. GRE alone doesn't offer security features.

#### **\*NAT-T\***

NAT-T (Network Address Translation-Traversal) allows IPSEC to work through NAT devices. NAT-T negotiation occurs in the IKE phase, specifically in the first two messages. It uses port 4500 UDP.

#### **\*Ports\*:**

- \*HTTP\*: 80
- \*HTTPS\*: 443
- \*DNS\*: 53
- \*DHCP\*: 67 (server), 68 (client)
- \*FTP\*: 21
- \*SSH\*: 22
- \*NAT-T\*: 4500 (UDP)

#### **\*Troubleshooting Half-Loaded Page\***

(when did you notice the issue ? , is it happening for all the URL or single URL ? ,Have you tried it in a different browser ? )

Check network connectivity with ping, verify DNS resolution with nslookup, examine browser console for errors, and check firewall settings that might block resources.

#### **\*Commands for Network Connectivity & DNS Resolution\*:**

Use ping for connectivity, tracert for route tracing, and nslookup or dig for DNS resolution to test name resolution.

#### **\*Zscaler\*:**

Zscaler is a cloud security platform providing secure internet access, data protection, and user security without the need for traditional on-premise appliances, leveraging a global network for policy enforcement.

#### **. \*Wireshark Filters\***

- \*SYN Packet\*: tcp.flags.syn == 1 && tcp.flags.ack == 0
- \*Filter Specific Website\*: frame contains "example.com"
- \*Filter by IP Address\*: ip.addr == ip address
- \*HTTP Requests\*: http.request